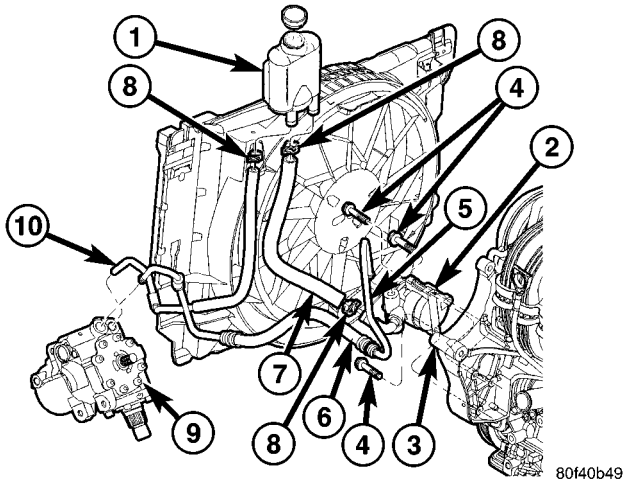

Fig. 3 POWER STEERING GEAR & PUMP - 2.4L

1 - REMOTE RESERVOIR
2 - POWER STEERING PUMP
3 - PUMP MOUNT BRACKET
4 - MOUNTING BOLTS (3)
5 - PRESSURE SWITCH LOCATION
6 - PRESSURE OUTLET HOSE (GEAR TO PUMP)
7 - RETURN HOSE (RESERVOIR TO PUMP)
8 - CLAMPS
9 - STEERING GEAR
10 - INLET HOSE (RESERVOIR TO GEAR)

OPERATION

The gear acts as a rolling thread between the worm shaft and rack piston. The worm shaft is supported by a thrust bearing at the upper end. When the worm shaft is turned from input from the steering column the rack piston moves. The rack piston teeth mesh with the pitman shaft. Turning the worm shaft turns the pitman shaft, which turns the steering linkage.

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - POWER STEERING SYSTEM DIAGNOSIS CHARTS

There is some noise in all power steering systems. One of the most common is a hissing sound evident at a standstill parking. Or when the steering wheel is at the end of it's travel. Hiss is a high frequency noise similar to that of a water tap being closed slowly. The noise is present in all valves that have a high velocity fluid passing through an orifice. There is no relationship between this noise and steering performance.